Sebastian Herrera Claro

DATA SCIENTIST · SYSTEMS ENGINEER

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SUMMARY

Systems engineer focused on data science. I build end-to-end projects on live public data, from API ingestion and cleaning to modeling, honest evaluation and interactive dashboards. My work covers classification, time-series forecasting and A/B testing / causal inference, and my software-development background helps me build reproducible pipelines. I care about leakage-free features, out-of-time validation, and recommendations a business can act on, including "don't deploy the model" when the data says so.

DATA SCIENCE PROJECTS

Email Campaign A/B Test & Uplift Modeling

Dashboard · Code

Python · SciPy · statsmodels · scikit-learn

- Analyzed a 64K-customer, three-arm randomized email experiment: checked the randomization (group sizes and covariate balance), then ran bootstrap confidence intervals, Holm multiple-testing correction, CUPED-style variance reduction and power analysis.
- Showed the men's email lifted visits **72%** and earned **$131 per 1,000 sends (95% CI $46–$216)**. Found the women's email works only on women's-line buyers.
- Compared S-, T- and X-learner uplift models over 20 resampled splits plus offline policy evaluation, and recommended **not** deploying model targeting because it gave no profit gain over a simple rule.

Great Britain Electricity Demand Forecasting

Dashboard · Code

Python · pandas · scikit-learn · NESO & Open-Meteo APIs

- Forecast half-hourly national electricity demand one day ahead with **4.2% MAPE**, half the error of a seasonal baseline, using archived weather *forecasts* and an automated leakage check.
- Benchmarked against the grid operator's official forecasts over 463 unseen days. The model lost on peak demand (802 vs. 578 MW error), but a fixed 50/50 blend **beat the operator's forecast by 6.7%** because the two forecasts err in opposite directions.
- Calibrated 80% prediction intervals with conformal adjustment, raising empirical coverage from 59% to 79%.

NYC 311 Service Delay Prediction

Dashboard · Code

Python · pandas · scikit-learn · Socrata API

- Built an end-to-end pipeline on ~400K NYC service requests pulled from the city's live open-data API, with logged data-quality rules and features restricted to information available at filing time.
- Found that a 0.95-ROC-AUC model was barely better than a lookup table on complaint type, then reframed the target to "slower than similar requests." On that target the model reaches **2.2× lift**: 64% of its top-10% flags are delayed, against a 29% base rate.
- Validated on a held-out future month and published an interactive dashboard that explains a time-of-day confounder in the data.

SKILLS

Languages & data	Python (pandas, NumPy), SQL (PostgreSQL, MySQL), MongoDB, REST APIs
Machine learning	scikit-learn, gradient boosting, regularized regression, uplift models (S/T/X-learners), feature engineering, time-based validation
Statistics	SciPy, statsmodels, A/B testing, bootstrap, multiple-testing correction, power analysis, CUPED, prediction intervals
Visualization	matplotlib, interactive SVG/HTML dashboards
Engineering	Git, GitHub Pages, JavaScript, React, Node.js, Java, C++

EDUCATION

Systems Engineering · Universidad de Cartagena — Mar 2019 – Dec 2024